

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 7: Primary Pulse Radar & Airborne Weather Radar (AWR)

Syllabus Source: CAE Oxford (pp. 253-296) | Pass Target: $\geq 90\%$ AviationExam

1. PRIMARY RADAR EQUATIONS & PRINCIPLES

Primary radar relies on the reflection of transmitted radio frequency pulses by a target:

- Pulse Recurrence Frequency (PRF): Number of pulses transmitted per second. Maximum Unambiguous Range: $R_{\max} = c / (2 \times \text{PRF})$. E.g. $\text{PRF} = 500 \text{ Hz} \rightarrow R_{\max} = 300,000 / 1,000 = 300 \text{ km}$.
- Pulse Length (tau): Duration of the pulse (microseconds). Minimum Blind Range: $R_{\min} = (c \times \tau) / 2$. Range Resolution = $(c \times \tau) / 2$.
- Beam Width & Angular Resolution: Beam Width = $(70 \times \lambda) / \text{Antenna Diameter}$. Angular resolution improves with a larger antenna and shorter wavelength!

2. AIRBORNE WEATHER RADAR (AWR) OPERATIONS

AWR operates in the SHF X-band at $\sim 9,375 \text{ MHz}$ (wavelength $\lambda = 3.2 \text{ cm}$), calibrated to reflect off wet precipitation droplets:

- Color Coding: Green (Light rain), Yellow (Moderate rain), Red (Heavy rain / CB core), Magenta (Severe turbulence detected by Doppler shift).
- Radar Shadow (Attenuation): A massive storm cell completely absorbs the radar beam. Behind the red core, a black shadow appears on the display. NEVER FLY INTO A RADAR SHADOW: A lethal second storm is hiding inside!
- Tilt Formula: Antenna Tilt (degrees) = $(\text{Height Difference in feet} / \text{Distance in NM} \times 100) - (\text{Beam Width} / 2)$.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Weather Radar Tilt Calculation &

Problem: Cruising at FL 300 (30,000 ft). Antenna beamwidth is 4.0° . Scan a storm cell at 40 NM without ground clutter. Step 1: Half beamwidth = $4.0^\circ / 2 = 2.0^\circ$. Step 2: Depression angle to ground at 40 NM: $\text{Height (NM)} / \text{Dist (NM)} \times 60 = (5 / 40) \times 60 = 7.5^\circ$ depression. Step 3: Tilt setting = $-(\text{Depression} - \text{Half Beamwidth}) = -(7.5^\circ - 2.0^\circ) = -5.5^\circ$. Operational Rule in High Cruise: Keep tilt around -1° to -2° to scan storm cores without ground reflection.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating AWR Antenna Tilt Setting

SCENARIO (Weather Radar Tilt Drill): An airliner is cruising at FL 330 (33,000 ft): -

A severe storm cell is detected 40 NM ahead - The freezing level (cloud base of hazard) is at 13,000 ft MSL - The radar antenna has a 4.0° beam width QUESTION: What antenna TILT angle must the pilot select so the BOTTOM of the radar beam sweeps through the 13,000 ft level at 40 NM? SOLUTION PASO A PASO (FOR DUMMIES): Step 1:

Calculate the Height Difference between aircraft and target level: - Aircraft

Altitude = 33,000 ft - Target Altitude = 13,000 ft - Height Difference = 33,000 ft

- 13,000 ft = 20,000 ft (target is BELOW aircraft). Step 2: Calculate the Depression

Angle (degrees) to the target level: - Rule of Thumb: 1° = 100 ft per Nautical

Mile! - Depression Angle = Height Difference / (Distance x 100) - Depression Angle

= 20,000 ft / (40 NM x 100) = 20,000 / 4,000 = 5.0° DOWN (-5.0°). Step 3: Adjust for

Half the Beam Width (Beam Width = 4.0°): - Beam Width / 2 = 4.0° / 2 = 2.0°. - To

place the BOTTOM of the beam at the target, the center of the beam must aim 2.0°

HIGHER: - Tilt Setting = Depression Angle + Half Beam Width = -5.0° + 2.0° = -3.0°

(3° DOWN)! FINAL ANSWER: Set antenna tilt to -3.0° DOWN.

[!] EXAM TRAP (AVIATIONEXAM) - The Radar Shadow Trap

- If a storm displays a deep black notch or blind zone directly behind a red cell: -

That black area is NOT clear air! It is a RADAR SHADOW caused by beam attenuation! -

Flying into a radar shadow is one of the most fatal thunderstorm traps in aviation.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What wavelength is used by standard Airborne Weather Radar (AWR) systems operating in the X-band?**

- [A]: 10 meters
- [B]: APPROXIMATELY 3.2 CENTIMETERS (frequency ~9,375 MHz)
- [C]: 10 cm
- [D]: 1 mm

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The 3 cm X-band provides optimal reflectivity from liquid raindrops while maintaining an antenna dish size that fits within an aircraft nose radome.

Q2: On an Airborne Weather Radar display, what does a completely BLACK area situated immediately behind a

- [A]: A cloud-free valley
- [B]: A RADAR SHADOW caused by heavy rain completely absorbing and attenuating the radar beam
- [C]: Safe flying zone
- [D]: Ground return

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The radar pulse cannot penetrate through dense water cores, creating a blind shadow where severe convective storms remain hidden.

Q3: What radar parameter determines the MINIMUM BLIND RANGE of a primary pulse radar?

- [A]: Antenna rotation speed

- [B]: PULSE LENGTH (tau) (the receiver cannot receive while the transmitter is transmitting)
- [C]: PRF only
- [D]: Wavelength

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. During pulse transmission (pulse duration tau), the receiver is disconnected to prevent damage: $R_{\min} = (c \times \tau) / 2$.